International University — VNUHCM
School of Computer Science and Engineering

MIDTERM EXAMINATION
Handwritten notes/notebooks are allowed; laptops/cellphones are not allowed.
April 05, 2022
(SET01)

TIME ALLOWED: 90 MINUTES

SUBJECT: COMPUTER ARCHITECHTURE (IT089IU)	
Approval by School of Computer Science and Engineering Signature Full name:	**Lecturer** Signature Full name: Le Hai Duong
Proctor 1 Signature Full name:	**Proctor 2** Signature Full name:

STUDENT INFO
Student name: Student ID:

INSTRUCTIONS TO STUDENTS

1. Purpose:
   - Evaluate students in understanding the principles of computer architecture and the interfaces between its hardware and software components; processes and process management (CLO1)
   - Evaluate students the ability to Create assembly programs and their machine code equivalent (CLO03)
   - Test ability to Analyze the performance of computer (CLO4)
2. Requirements:
   - Give explaination for all the your results.
   - Keep your eyes on your own paper.

Table 1: CPI classes

Processor	Clock rate	Class A	ClassB	ClassC	Class D
P1	2.5GHz	3	3	1	2
P2	3GHZ	2	1	2	3

QUESTIONS

1. **(30 marks)** Consider two different implementations (the processor P1 and P2) of *the same instruction set architecture*. The instructions can be divided into four classes according to their CPI (class A, B, C, and D) as shown in **Table 1.**
   Given a program with a dynamic instruction count of 10,000,000 instructions divided into classes as follows: 20% class A, 10% class B, 40% class C, and 30% class D.

   a. Find the clock cycles required for the program in both implementations. **(10 marks)**
   b. What is the average CPI for each implementation? **(10 marks)**
   c. Which implementation is faster? **(10 marks)**

2. **(20 marks)** For the following C codes, write the corresponding MIPS assembly codes? Assume that the variables i, j and k are assigned to registers $\$s0$, $\$s1$ and $\$s2$, respectively; the base address of the array A is in register $\$s3$.
   a. **Code 1:** **(10 marks)**

```
if (i == j)
    A[i] = A[k] + 1;
else
    A[i] = A[j] - 1;
```

   b. **Code 2:** **(10 marks)**

```
while (i != j) {
    i += 1;
    A[k - i] = 0;
}
```

3. **(40 marks)** Assume that the C-level integer i is held in register $\$t1$, $\$s2$ holds the C-level integer called *result*, and $\$s0$ holds the base address of the integer $MemArray$. Given the following assembly:

```
(1)          .data
(2)  MemArray:   .word  1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

(3)          .text
(4)          .globl main
(5)  main: addi    $t1, $0, 1
(6)        add     $s2, $0, $0
(7)        la      $s0, MemArray

(8)  LOOP: lw      $s1, 0($s0)
(9)        addi    $s0, $s0, 4
(10)       addi    $t1, $t1, 2
(11)       add     $s2, $s2, $s1
(12)       slti    $t2, $t1, 9
(13)       bne     $t2, $0, LOOP
```

```
(14)       move    $a0, $s2
(15)       li      $v0, 1       # syscall(1) to print a integer
(16)       syscall

(17)       li $v0, 10
(18)       syscall
```

Answer the following questions with explanation if applicable.

a. Write comments for the given assembly. **(5 marks)**
b. What is the value of $\$t1$ at Line 10 after the **first** round of the $LOOP$? **(5 marks)**
c. What is the value of *result* after the **first** round of the $LOOP$? **(5 marks)**
d. How many times does the $LOOP$ execute? **(5 marks)**
e. What is the value of i after the $LOOP$ finished? **(5 marks)**
f. What is the value of *result* after the $LOOP$ finished? **(5 marks)**
g. What character does the assembly print out? **(5 marks)**
h. Convert the *main* procedure into C-language code. **(5 marks)**

4. **(5 marks)** Assume the following register contents: $\$t0 = 0xAAAAAAAA$, $\$t1 = 0x01020304$. For the register values shown above, what is the value of $\$t2$ for the following sequence of instructions?

```
sll    $t2, $t0, 28
or      $t2, $t2, $t1
andi    $t2, $t2, -1
```

5. **(5 marks)** Suppose that the values in the registers $\$t1$, $\$t2$ are $0xDEADBEEF$ and $0x00000000$, respectively. What is the bit pattern at the memory address $0x00000002$ after the following instruction?

```
sw $t1, 0($t2)
```

Table 2: CPI classes

Processor	Clock rate	Class A	ClassB	ClassC	Class D
P1	2.5GHz	3	3	1	2
P2	3GHZ	2	1	2	3

Answer Key for Exam A

QUESTIONS

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   Given a program with a dynamic instruction count of 10,000,000 instructions divided into classes as follows: 20% class A, 10% class B, 40% class C, and 30% class D.

   a. Find the clock cycles required for the program in both implementations. **(10 marks)**
   b. What is the average CPI for each implementation? **(10 marks)**
   c. Which implementation is faster? **(10 marks)**

   Answer: a. $clock\ cycles\ P1 = \sum(CPI \times I)$
   $= 3 \times 2 \times 10^6 + 3 \times 10^6 + 1 \times 4 \times 10^6 + 2 \times 3 \times 10^6 = 19 \times 10^6$
   $clock\ cycles\ P2 = 2 \times 2 \times 10^6 + 1 \times 1 \times 10^6 + 2 \times 4 \times 10^6 + 3 \times 3 \times 10^6 = 22 \times 10^6$
   b. $Avg.\ CPI\ P1 = cycles/ins = 19 \times 10^6/10 \times 10^6 = 1.9$

   $Avg.\ CPI\ P2 = 2.2$

   c. $CPU\ time\ P1 = cycles/clockrate = 19 \times 10^6/2.5 \times 10^9 = 7,6 \times 10^{-3}s$

   $CPU\ time\ P2 = 22 \times 10^6/3 \times 10^9 = 7.33 \times 10^{-3}$

   P2 is faster than P1.

2. **(20 marks)** For the following C codes, write the corresponding MIPS assembly codes? Assume that the variables i, j and k are assigned to registers $\$s0$, $\$s1$ and $\$s2$, respectively; the base address of the array A is in register $\$s3$.
   a. **Code 1:** **(10 marks)**

```
if (i == j)
    A[i] = A[k] + 1;
else
    A[i] = A[j] - 1;
```

   b. **Code 2:** **(10 marks)**

```
while (i != j) {
    i += 1;
    A[k - i] = 0;
}
```

   Answer: a.

```
    bne $s0, $s1, ELSE
    # load and compute A[k] + 1
    sll $t0, $s2, 2
    add $t0, $s3, $t0
    lw  $t0, 0($t0)
    addi $t0, $t0, 1

    # save to A[i]
    sll $t1, $s0, 2
    add $t1, $s3, $t1
    sw $t0, 0($t1)
    j EXIT
ELSE:
    # load and compute A[j] - 1
    sll $t0, $s1, 2
    add $t0, $s3, $t0
    lw  $t0, 0($t0)
    addi $t0, $t0, -1

    # save to A[i]
    sll $t1, $s0, 2
    add $t1, $s3, $t1
    sw $t0, 0($t1)
EXIT:
```

b.

```
LOOP:
    beq $s0, $s1, EXIT
    addi $s0, $s0, 1

    # A[k - i] = 0
    sub $t0, $s2, $s0
    sll $t0, $t0, 2
    add $t0, $s3, $t0
    sw  $zero, 0($t0)

    j LOOP
EXIT:
```

3. **(40 marks)** Assume that the C-level integer i is held in register $\$t1$, $\$s2$ holds the C-level integer called *result*, and $\$s0$ holds the base address of the integer $MemArray$. Given the following assembly:

```
(1)          .data
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(3)          .text
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(6)        add     $s2, $0, $0
(7)        la      $s0, MemArray
```

```
(8)  LOOP: lw      $s1, 0($s0)
(9)        addi    $s0, $s0, 4
(10)       addi    $t1, $t1, 2
(11)       add     $s2, $s2, $s1
(12)       slti    $t2, $t1, 9
(13)       bne     $t2, $0, LOOP

(14)       move    $a0, $s2
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```

Answer the following questions with explanation if applicable.

a. Write comments for the given assembly. **(5 marks)**
b. What is the value of $\$t1$ at Line 10 after the **first** round of the *LOOP*? **(5 marks)**
c. What is the value of *result* after the **first** round of the *LOOP*? **(5 marks)**
d. How many times does the *LOOP* execute? **(5 marks)**
e. What is the value of i after the *LOOP* finished? **(5 marks)**
f. What is the value of *result* after the *LOOP* finished? **(5 marks)**
g. What character does the assembly print out? **(5 marks)**
h. Convert the *main* procedure into C-language code. **(5 marks)**

Answer: b. $\$t1 = 3$
c. $\$s2 = result = 1$
d. Loop runs 4 times.
e. $\$i = 9$
f. $\$s2 = result = 10$
g. Print 10.
h.

```
#include <stdio.h>

int main()
{
    int MemArray[] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12};
    int i = 1;
    int j = 0;
    int result = 0;
    do {
        result += MemArray[j++];
        i += 2;
    } while (i < 9);

    printf("Result = %d, rounds =  %d, i =  %i", result, j, i);

    return 0;
}
```

4. **(5 marks)** Assume the following register contents: $\$t0 = 0xAAAAAAAA$, $\$t1 = 0x01020304$. For the register values shown above, what is the value of $\$t2$ for the following sequence of instructions?

```
sll    $t2, $t0, 28
or      $t2, $t2, $t1
andi    $t2, $t2, -1
```

Answer: a1020304

5. **(5 marks)** Suppose that the values in the registers $\$t1$, $\$t2$ are $0xDEADBEEF$ and $0x00000000$, respectively. What is the bit pattern at the memory address $0x00000002$ after the following instruction?

```
sw $t1, 0($t2)
```

Answer: 0xBE